

KINEMATIC CHALLENGE:

STCM Mechanical System.

Designer: Benoît Collin — Montreal, Québec.

License: Public Domain / Open Source

This document explains the primary reasons why so-called perpetual wheels fail and clearly points out their 3 main structural flaws.

And it is by targeting these 3 weaknesses in such an obvious manner that we now have the opportunity to finally bring concrete solutions to this never-resolved historical dilemma...

Obviously, theoretical science claims this is a violation of the laws of thermodynamics (which govern energy and heat losses), but the disconcerting simplicity of this work allows us to understand that the fundamental reasons for these failures are purely geometric and kinematic (the art of pure motion through geometry).

"THIS KINEMATIC JOURNEY exposes very clearly the three main problems of all known perpetual wheels, and showcases the difference in approach between wanting to just create perpetual motion and a perfectly kinematic geometric approach clearly designed to create driving force at the hub.

Theoretical science claims it is impossible; but some time ago, this very same science also said that matter, being heavier than air, could never

fly... The kinematic architecture I am about to reveal to you proves that yes, we can cause rotational imbalances on axles by moving the masses closer to or further away from the center.

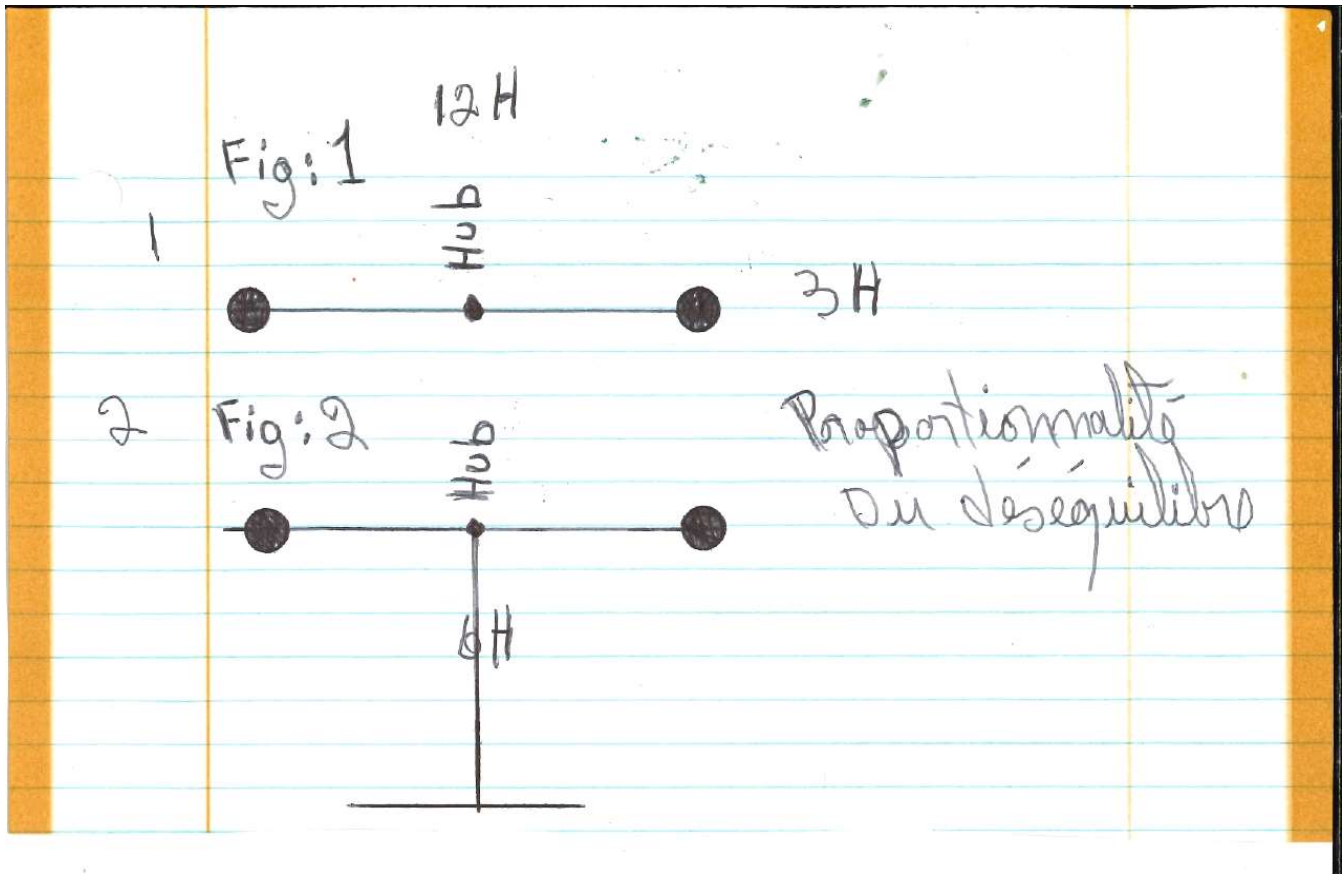
This is in no way magic, but rather the geometry of movements in perpetual imbalance, predetermined beforehand, as we will see throughout this work.

The primary reasons why all known perpetual wheels either do not work or fail to produce any useful torque at the hub are as follows:

The imbalance caused by moving the masses closer to or further away from the hub center is always too weak. To quickly understand what this involves, let us look closely at the nature of this imbalance:

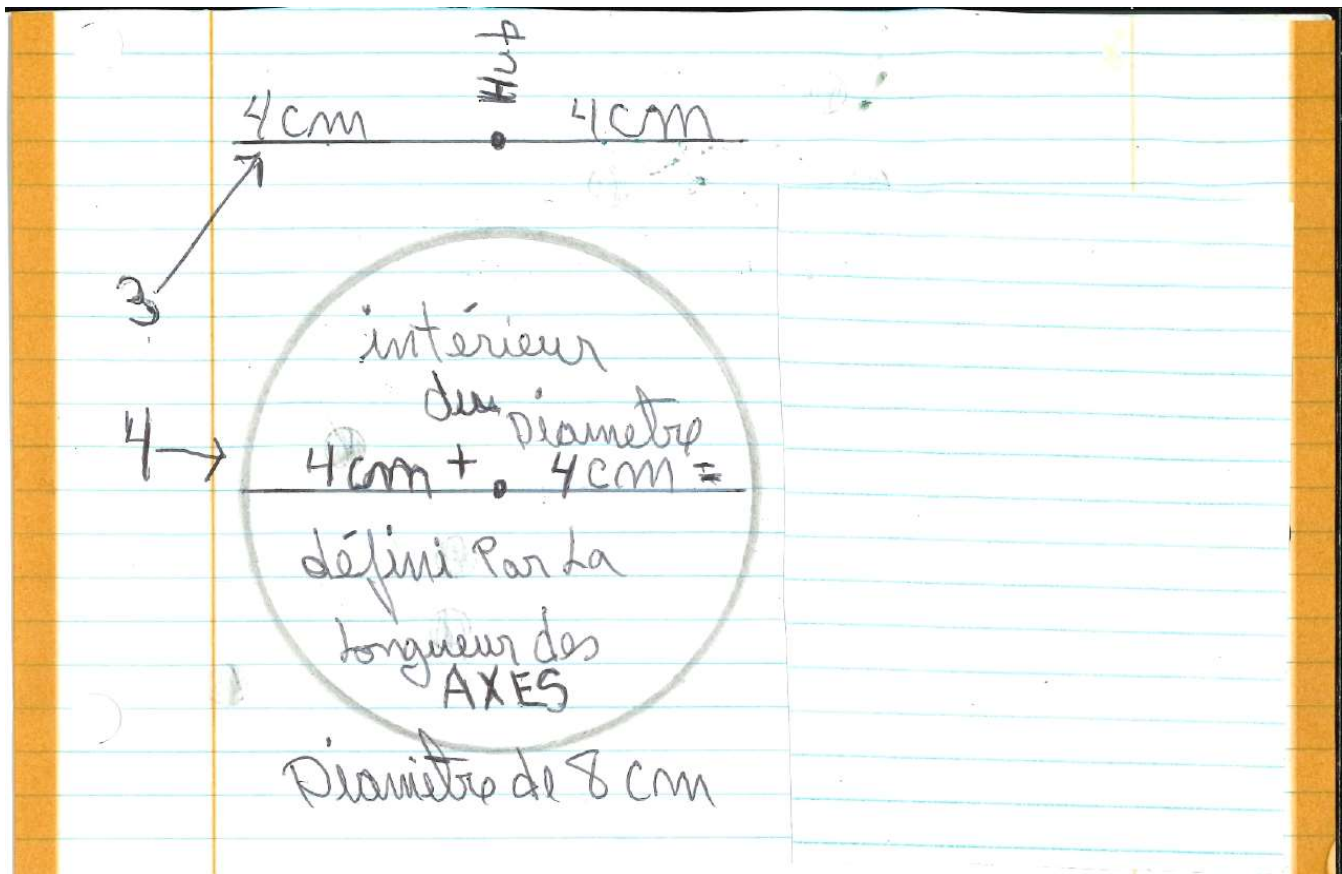
- Figure 1: Illustrates a system of masses in a state of perfect static balance.
- Figure 2: Introduces a slight imbalance, and it is precisely this overly slight imbalance that causes the problem when analyzing mass dynamics.

Known perpetual wheels rely on an even smaller proportion than Figure 2, which I purposely wanted to clearly distinguish from the balanced state in Figure 1 to demonstrate that it induces only a very weak imbalance.



Known perpetual wheels rely on an even smaller proportion than Figure 2, which I purposely wanted to clearly distinguish from the balanced state in Figure 1 to demonstrate that it induces only a very weak imbalance.

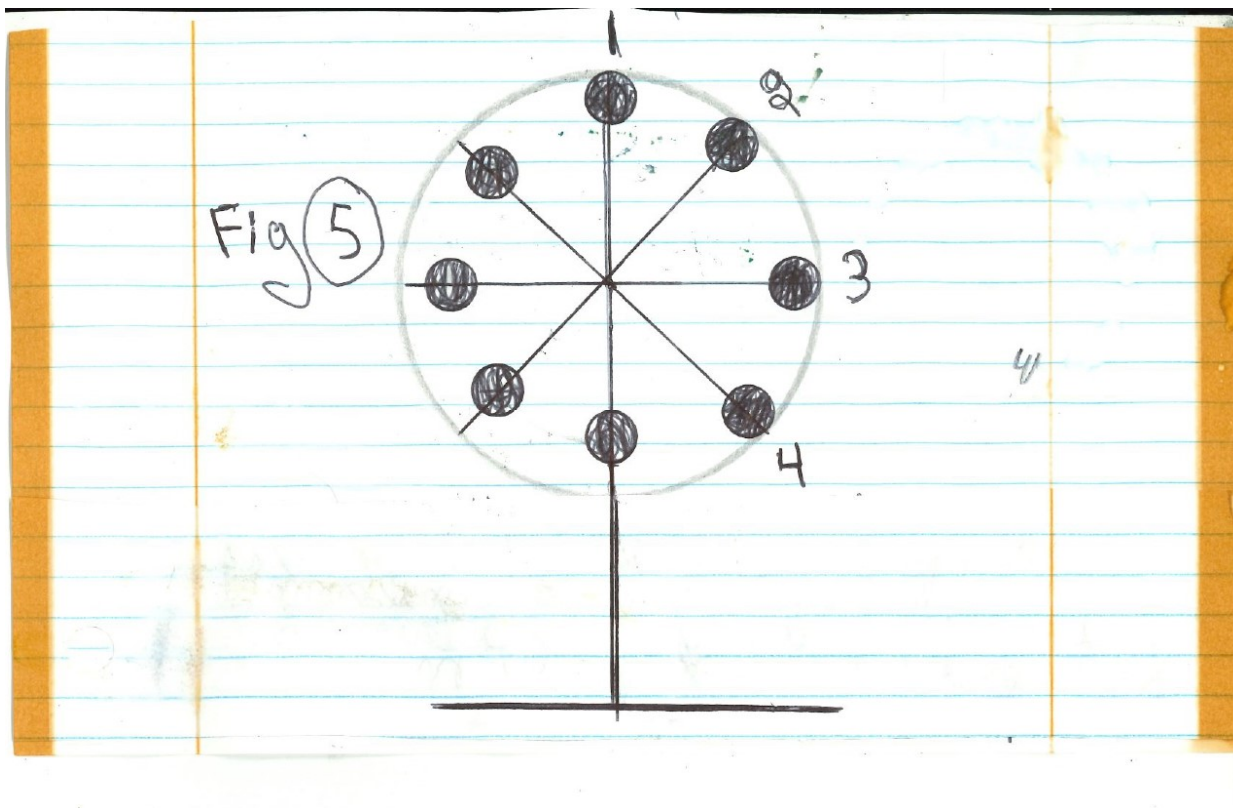
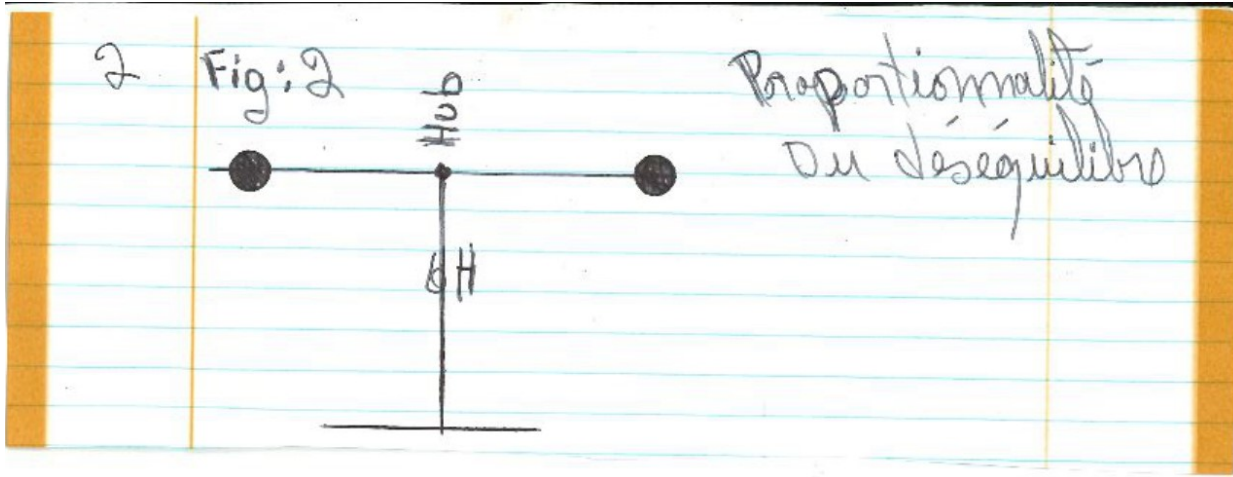
Figure 3: Shows an 8 cm axle. Figure 4: Shows the rotating diameter of the 8 cm axle.



Here below in Figure 5, we have reproduced the kinematics of Aldo Costa's wheel, as it already exists and is highly representative of existing kinematic approaches—which are far too weakly imbalanced.

For the sake of simplicity, I have limited this illustration to only 4 axles.

Here, we clearly see what this kinematic approach relies on to rotate this 4-axle structure (see the mechanical principle illustrated at the base of Figure 2).



Obviously, I have not illustrated the mechanisms that raise the masses

at 6 and 12 o'clock, which require energy to lift these two masses simultaneously, thereby causing significant resistance.

The real Aldo Costa wheel features 64 axles and 128 masses, which implies that there are still 31 heavier masses on the right providing support; however, such an approach with just 4 axles as drawn here would clearly not work.

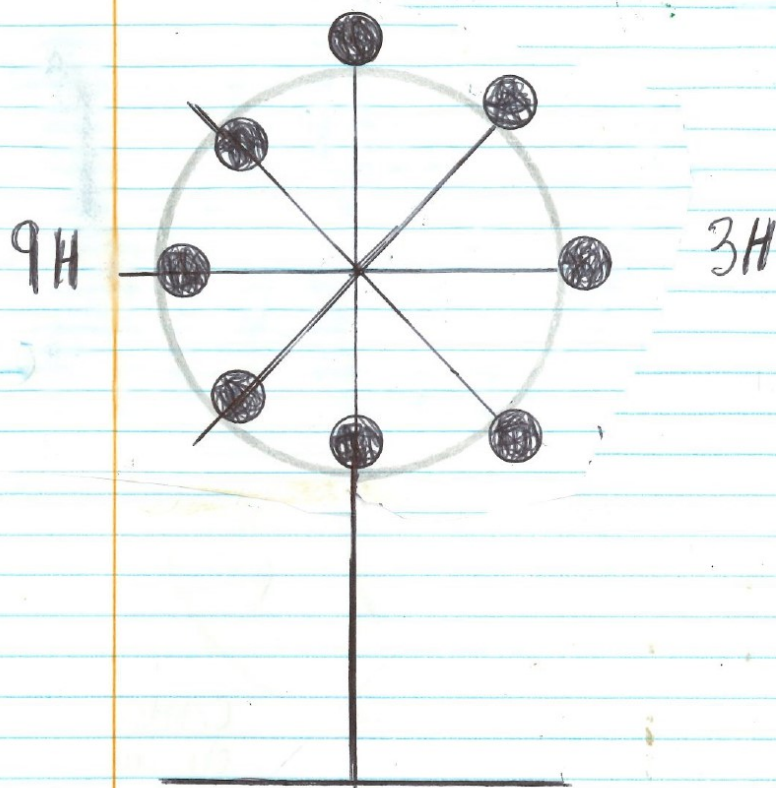
As we can see, the effect of the imbalance in this structure is understood without much explanation: it clearly rotates to the right.

The design is simple and clever, were it not for the dilemma encountered at 6 and 12 o'clock.

This structure would rotate and produce, in this specific proportion of imbalance, a weak torque at the center.

Still keeping 4 axles, in Figure 5.5, we improve Aldo Costa's kinematics.

Fig. 5.5



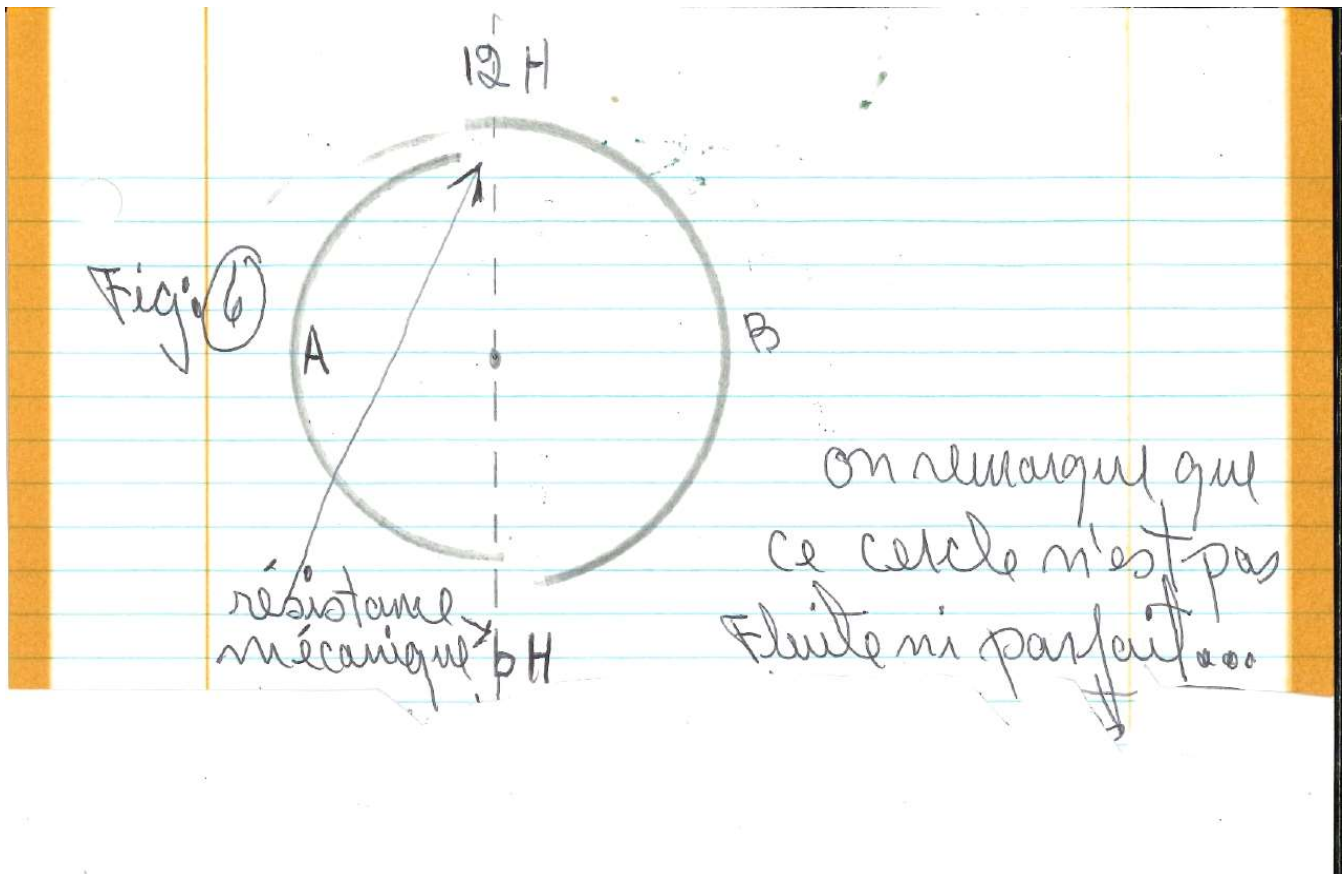
hus, even with the dilemma poorly resolved at 6 and 12 o'clock, Aldo Costa would have had a weakly driving wheel instead of just a capriciously perpetual wheel.

Notice that I had to make our descending masses on the right exit the centered diameter relative to the hub, and that I had to adjust the geometry by lengthening the 3 axles on the left...

which implies that the right axles would have also been lengthened just like the ones on the left.

Now let's look at the trajectory of the masses in perpetual wheels of the Aldo Costa type.

Here in Figure 6, the trajectory of the left masses and the trajectory of the right masses are traced, in reference to Figure 5 and to the main known perpetual wheels.



This kinematics remains simple to understand, but it clearly exposes the challenge posed at 6 and 12 o'clock by these two different diameters, which is still poorly resolved in known perpetual wheels.

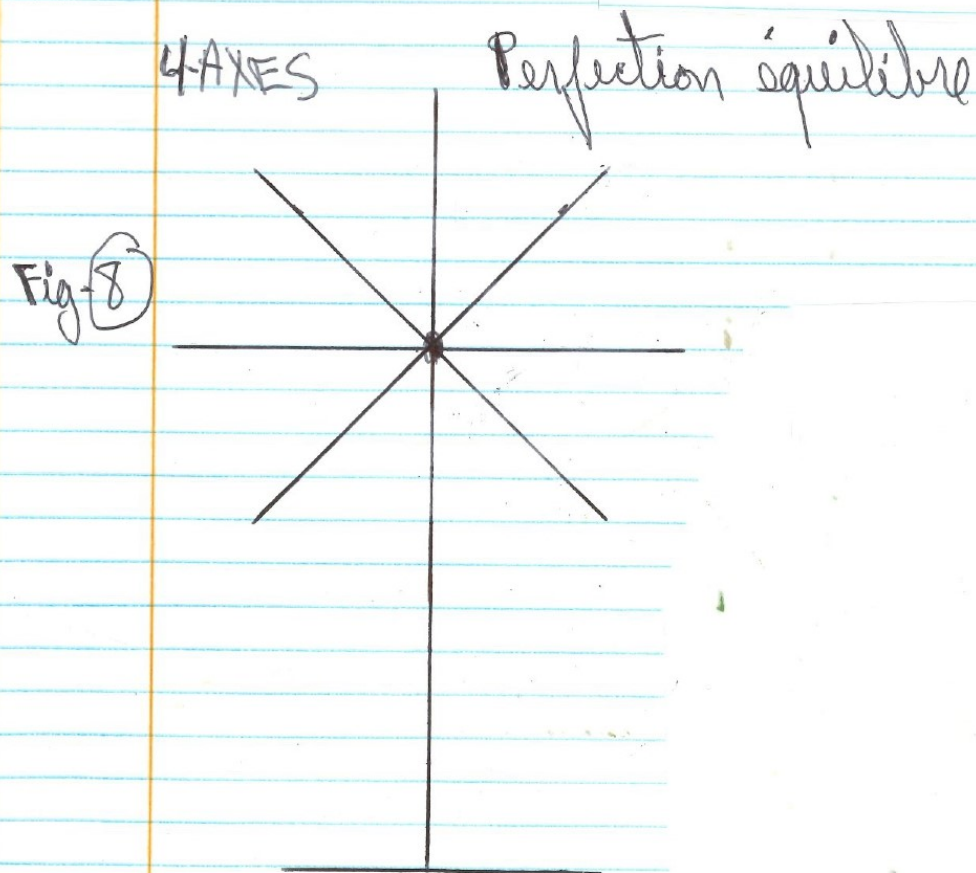
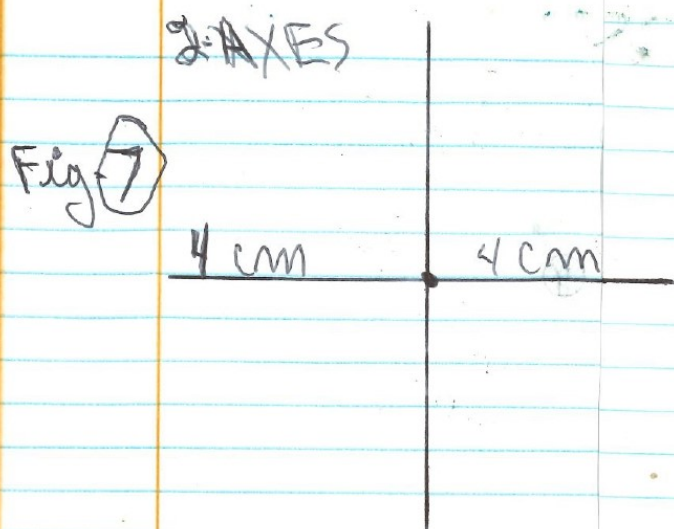
In Figure 7 below, we must understand that for a conventional perpetual wheel to work, the circle must be divided to absolute perfection, and truly nothing less.

If dividing a wheel by 2 is fairly easy, by 4 it is twice as difficult, by 8 even less so, and by 64 it is three times more complex than by 8.

With a diameter of 18 meters, like Aldo Costa's wheel, it becomes a

matter of high-precision craftsmanship.

It is highly unlikely that a human could achieve this feat without the help of precision tools on par with those used to manufacture satellites, because such a wheel will not be able to optimize the small gain pre-established at the kinematic start without perfect balance between all of its axles.



Now that we understand that the initial imbalance of known perpetual wheels is too weak to produce usable torque, the balance between each of the axles must be rigorously perfect—and that is without even mentioning the mechanical dilemma encountered at 6 and 12 o'clock.

I consider that Aldo Costa's wheel, despite its mixed success, remains highly instructive in the sense that it indicates an approach to creating a permanent imbalance inside a rotating structure.

However, while this approach is accurate for seeking motion, its kinematics strictly forbids any production of torque.

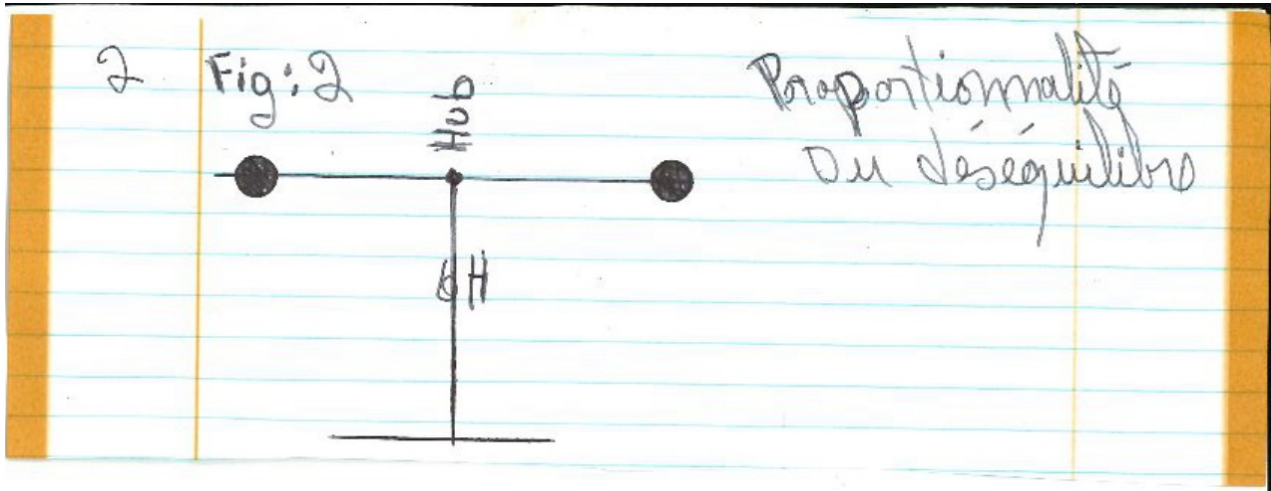
Even if such a wheel were executed successfully according to current approaches, it could only operate under a glass dome; because it produces no torque, it is only normal that a single minor resistance or friction would eventually bring it to a stop.

What this wheel teaches us, since we can identify its two main errors, is namely an initially induced imbalance that is far too weak, and the broken circular fluidity caused by the different diameters of the masses at 6 and 12 o'clock—where the masses ought to evolve smoothly and circularly without counterproductive tampering.

When we see the errors, do we not also partially see the solutions?

Observe the overly weak kinematic imbalance induced in known

wheels (see Fig. 2).



We can imagine the solutions as long as we properly identify the problems. Let's just think of a weighing scale, and let's rethink imbalance as it should first be understood, based on these three very simple-to-understand figures.

dessin du Volet no I Principe de base

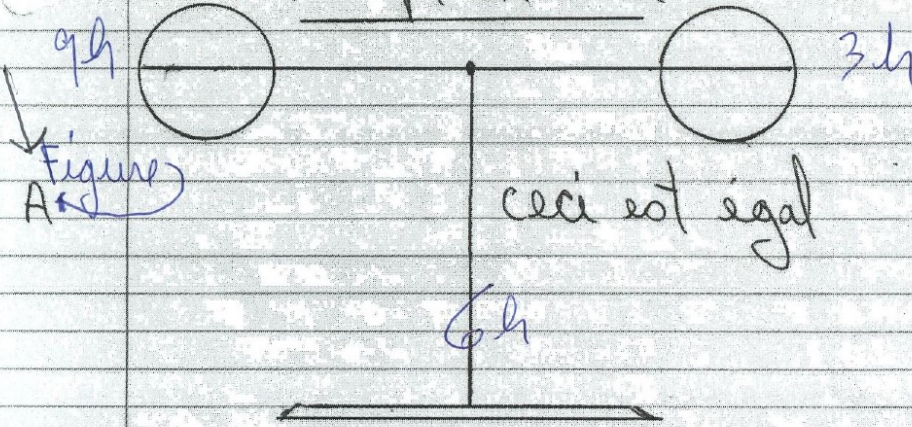


Figure A

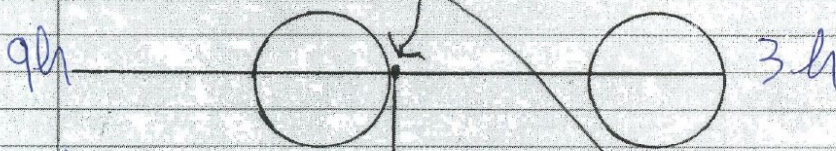
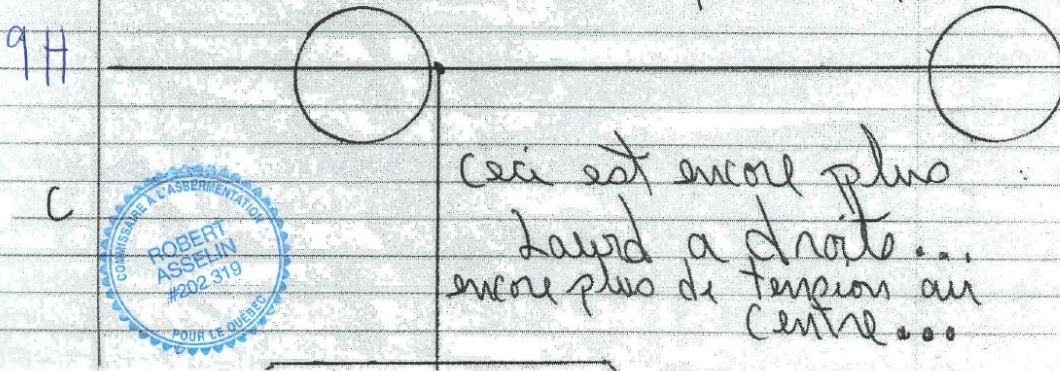


Figure B

ceci est plus lourd à droite...
Donc va exercer plus de tension au centre



C



As you can see here, this basic kinematics does not just graze the surface of imbalance like all known perpetual wheels.

Instead, it maximizes the imbalance to ensure that what goes down on the right is truly heavier than what comes back up on the left.

From positions B and C, we can notice that the left mass is very close to the hub and the right mass is furthest from the central hub at the 9 and 3 o'clock positions.

This starting kinematics has nothing to do with known perpetual wheels.

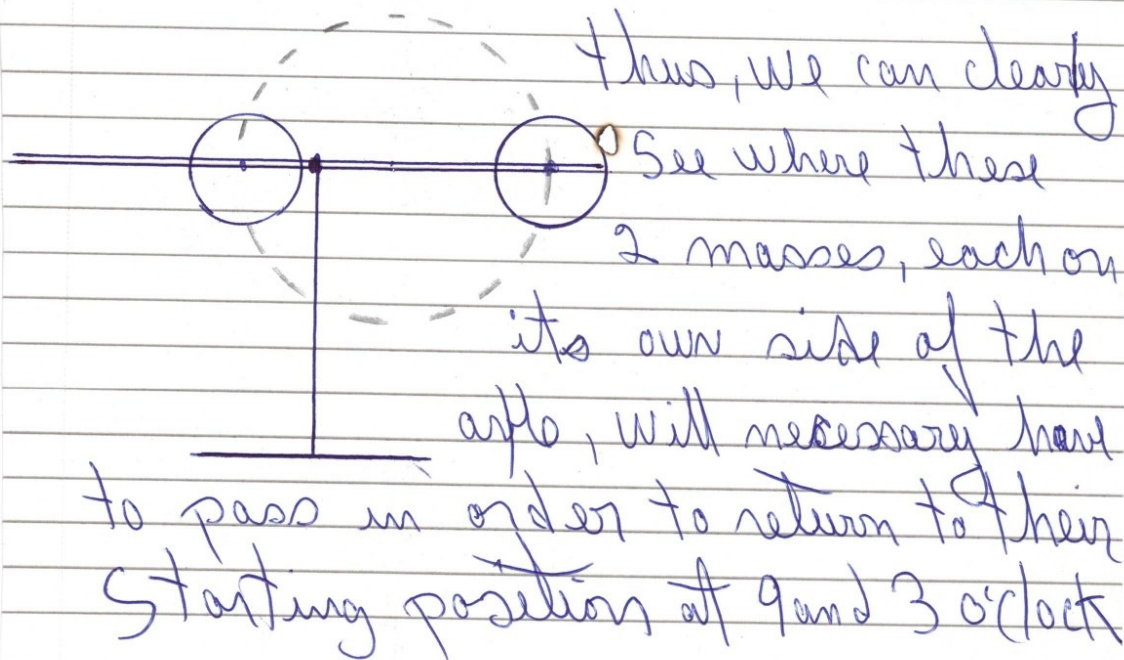
This geometric configuration aims to generate genuine driving force at the center hub, rather than trying to create useless perpetual motion.

A rotating structure designed to turn more heavily on the right than on the left requires, as traditional systems confirm, that what goes down on the right is substantially heavier than what goes up on the left.

This depends mainly on the distance of the masses relative to the hub. If the goal is truly to create torque, right from the start of the rotation, we must bring the left mass very close to the hub, as illustrated in the previous drawings.

page (2)

Based on the foregoing, if we were to perform a complete 360° rotation starting from the Horizontal position, that is, the 9 and 3 o'clock position, so that these masses return exactly to their starting position of 9 and 3 o'clock after their complete revolution, we would proceed as follows:



How is this achieved?

Take the centers of your two masses exactly at the horizontal 9 and 3 o'clock positions, measure that distance, and divide it by two.

Place your compass at the center, still on this 9 and 3 o'clock line, and close the circle onto both centers.

This shows you exactly where these masses must kinematically pass in order to complete a full revolution and return to their exact same starting position.

In this kinematics, a real and blatant imbalance is evident right from the start, where everything going down on the right is not of an equivalent weight to the left, but is significantly heavier than what goes up.

Here, a very clear difference in physical intent is visible through the lever arm, particularly at 9 and 3 o'clock.

By analyzing this dynamic, it is evident that we are not searching for useless perpetual motion, but are truly looking to create a highly usable driving force at the center hub.

What is a driving wheel ?

It is a system where everything going down is, or becomes, considerably heavier than everything going up on the left. Thus, driving force is obtained at the hub through the internal and perfectly circular

management of its own weight.

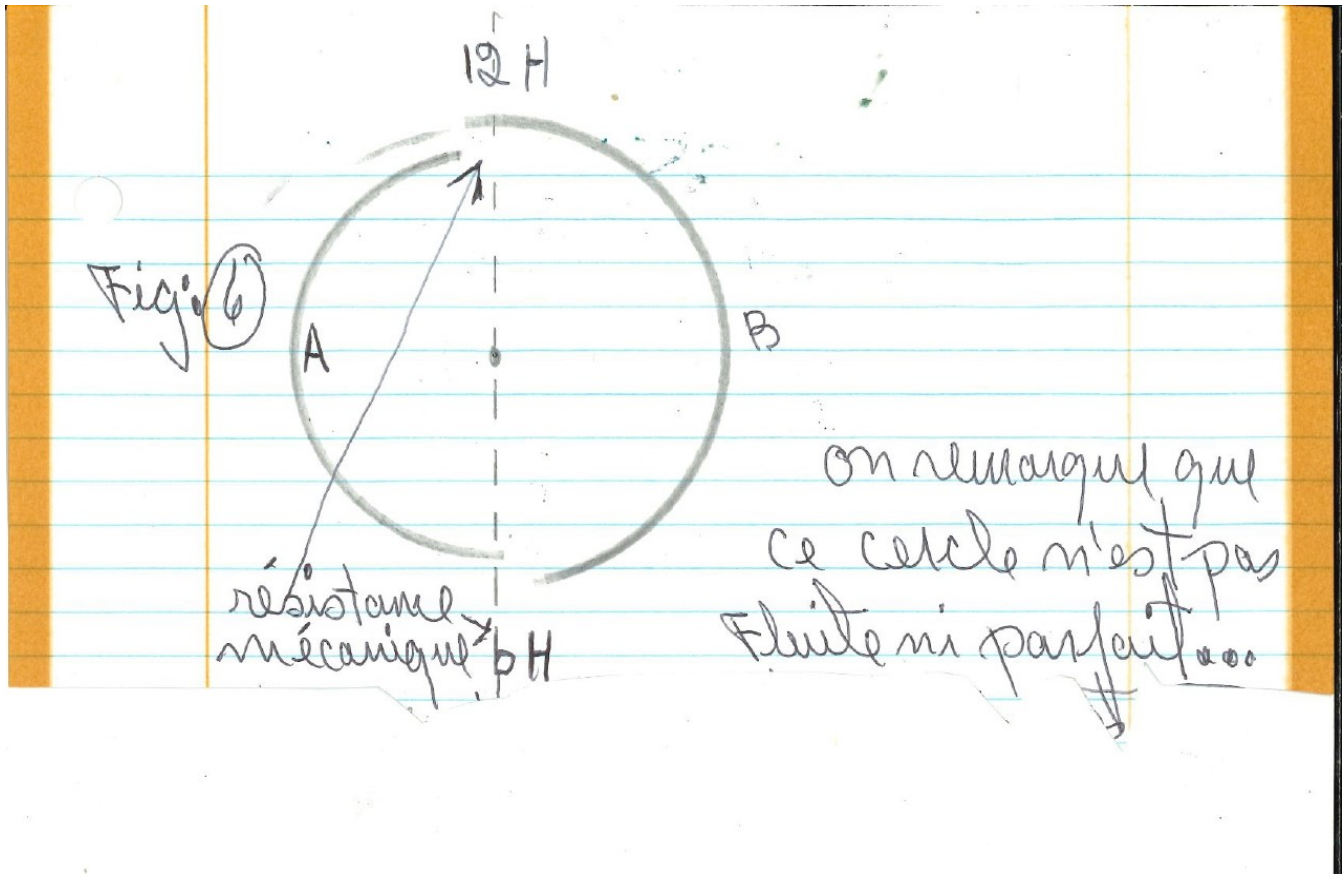
What is a perpetual wheel?

It is a structure where the initial imbalance at the base is far too weak, and where the rotating trajectory of the masses breaks at 6 and 12 o'clock into two different diameters.

This causes the system to either fail to work, or to operate under a glass dome without producing any torque beyond what is necessary for its own movement.

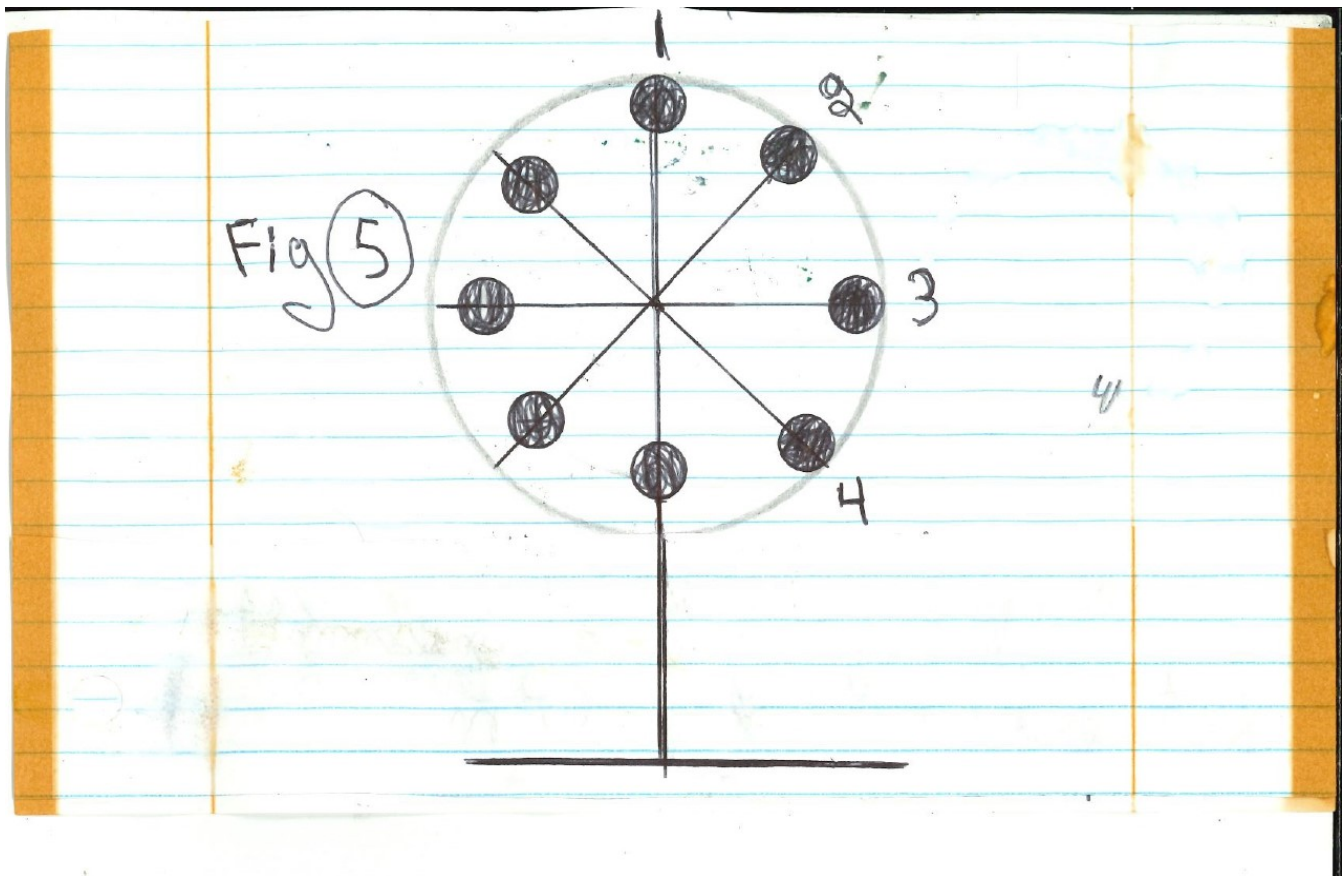
Consequently, these mechanical structures tolerate no friction and no resistance, which is exactly what defines a conventional perpetual wheel.

Returning to Figure 6, remember the two different diameters of the circular trajectory of the masses, broken at 6 and 12 o'clock in the previous Figures 5 and 5.5 due to the lack of a single, uniform diameter.

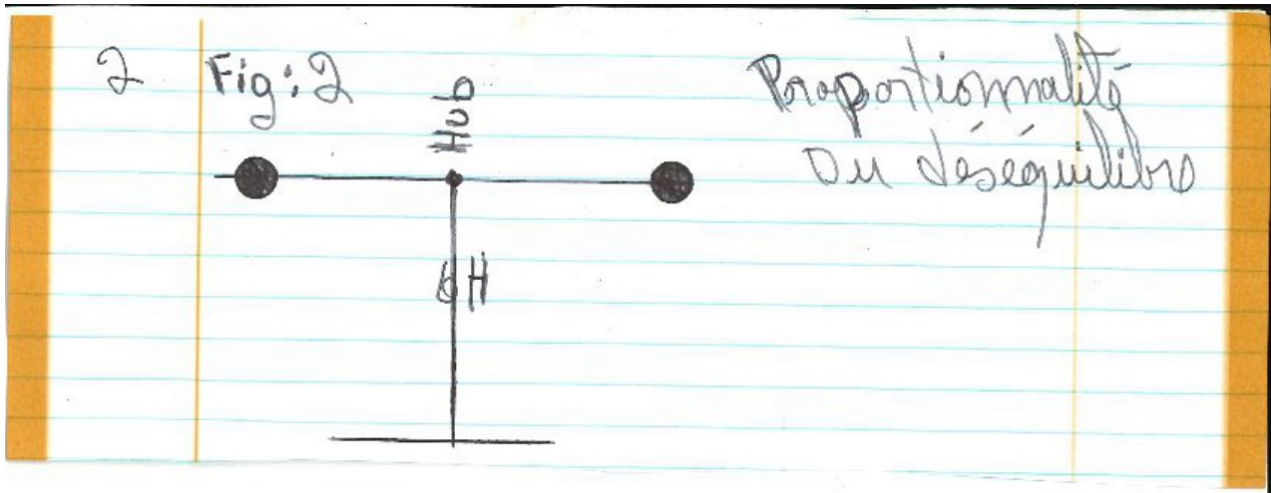


It is precisely this kinematics that is reflected by all known wheels, and very well represented by Aldo Costa's wheel, reduced here to only 4 axles in Fig. No. 5.

(Note that my explanations are by no means a direct criticism of Aldo Costa's work; on the contrary, this wheel provides us, simple as it looks, with the kinematic information on how to proceed better...)

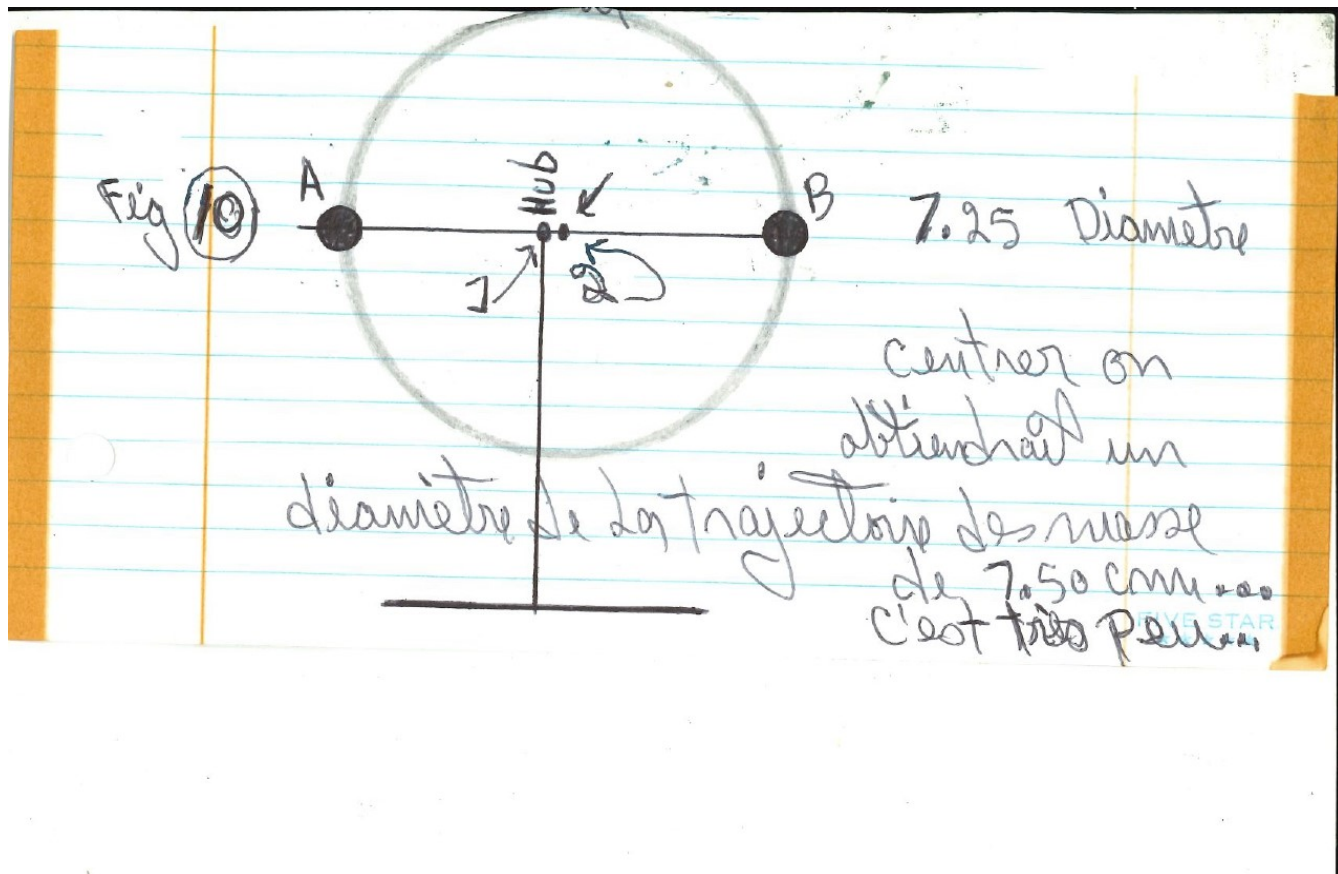


If, at its core, this kinematics in Figure 5 stems from this shown lower down in Figure 2:



We can take these same proportions and approach the dilemma differently than by adjusting the geometry poorly at 6 and 12 o'clock.

Let's see in the next Figure No. 10: by drawing a circle from the center of our two masses (where 1 is the central hub and 2 is the point of my compass to draw this circle) and by constraining the masses to move only within this circle, we obtain a perfect circle without the severe drawbacks at 6 and 12 o'clock found in known perpetual wheels.



Geometric Diameter Variations (Figures 11, 12, and 13) Viewed this way, since this circle is the mandatory path through which the masses must pass, the traditional kinematic and mechanical dilemma encountered at 6 and 12 o'clock is entirely eliminated.

Centered Configuration: The rotating diameter of the masses would be 7.50 cm. However, with this specific eccentric approach, our mass trajectory diameter is 7.25 cm; this kinematics produces torque, though weakly.

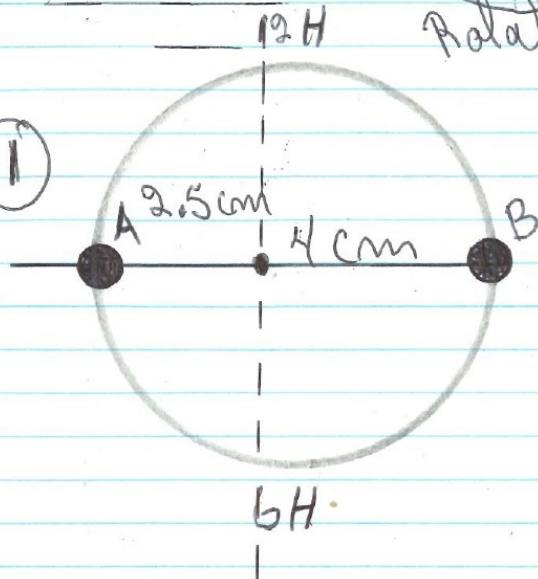
Figure 11: By bringing the left mass closer to the hub, we obtain a

rotating mass trajectory diameter of 6.5 cm.

Figure 12: By bringing mass A even closer to the hub, the diameter shifts to 5.6 cm.

Figure 13: The diameter is reduced down to 4.6 cm.

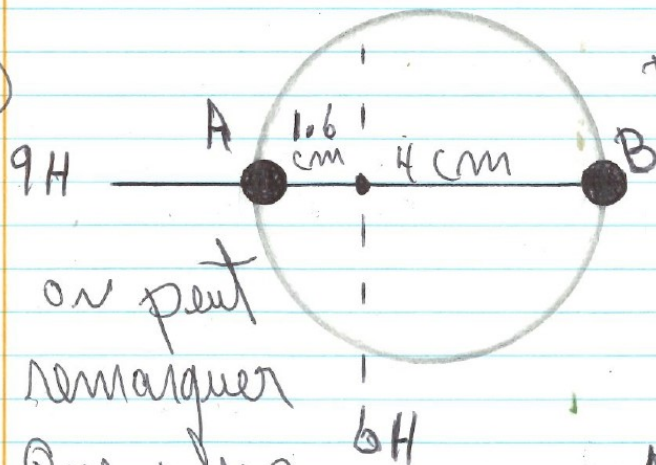
Fig: 11



ici le diamètre relatif des masses est de 6.5 cm

diamètre 6.5 cm

12



Diamètre Relatif 5.6 cm

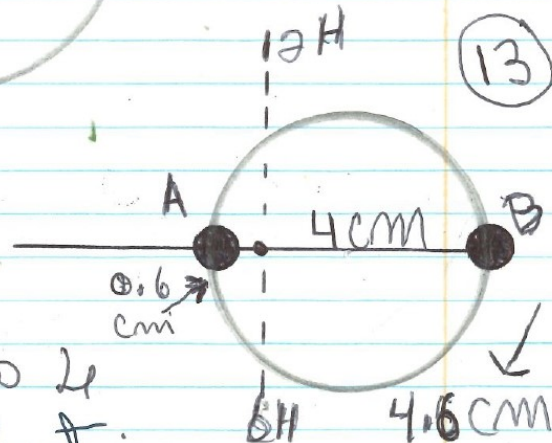
on peut remarquer que plus on rapproche

A du centre plus le

Diamètre de la trajectoire

Des masse se rapproche à gauche...

13



4.6 cm

Further down here, let's look again at our configuration with only one axle at the 9 and 3 o'clock position, and let's let it drop as shown in Figure 14, to realize that this axle alone and its two masses will not pass the 27-28 minute threshold—this is the kinematic dead center.

However, Figure 15 shows us how to solve this dilemma and easily cross that 28-minute threshold... watch closely what is about to happen:

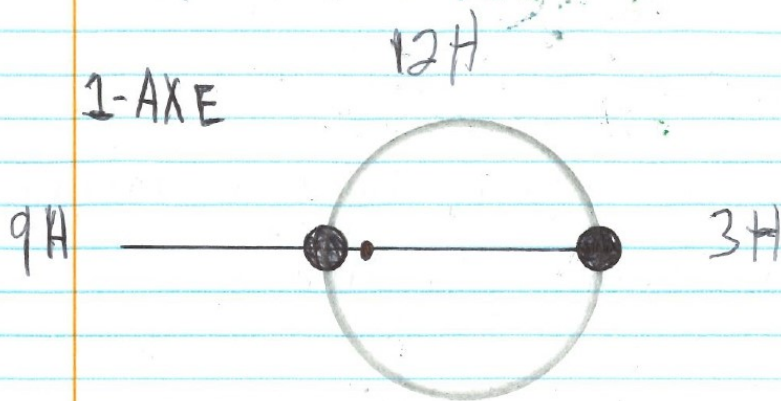
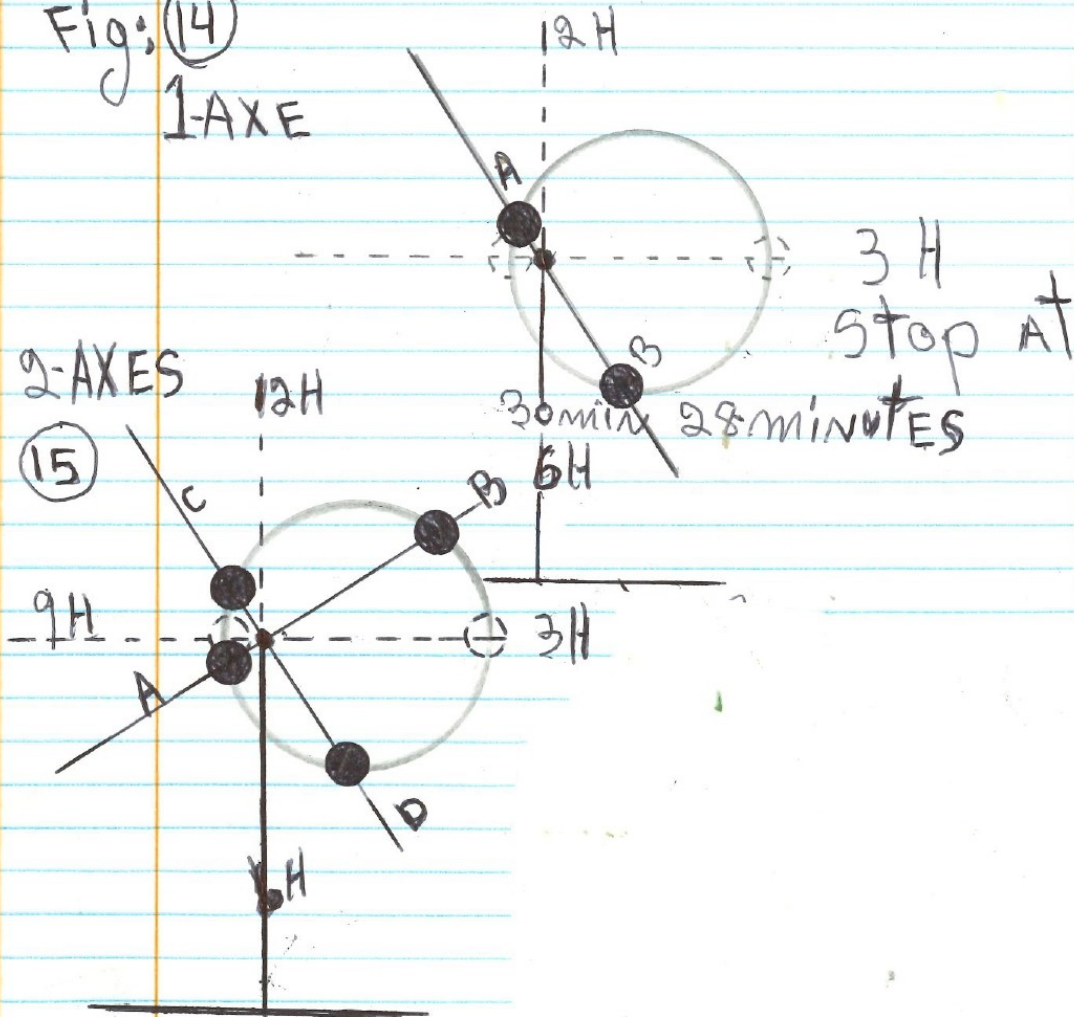


Fig: (14)
1-AXE



If in Figure 14 we see that A and B cannot go further than 28 minutes, we can clearly see in Figure 15 that we solve this dead center problem simply by adding a second axle positioned perpendicularly on the same shaft, as if, with two legs, it were much easier to move forward to travel.

I place great emphasis on this kinematics shown in Figure 15:



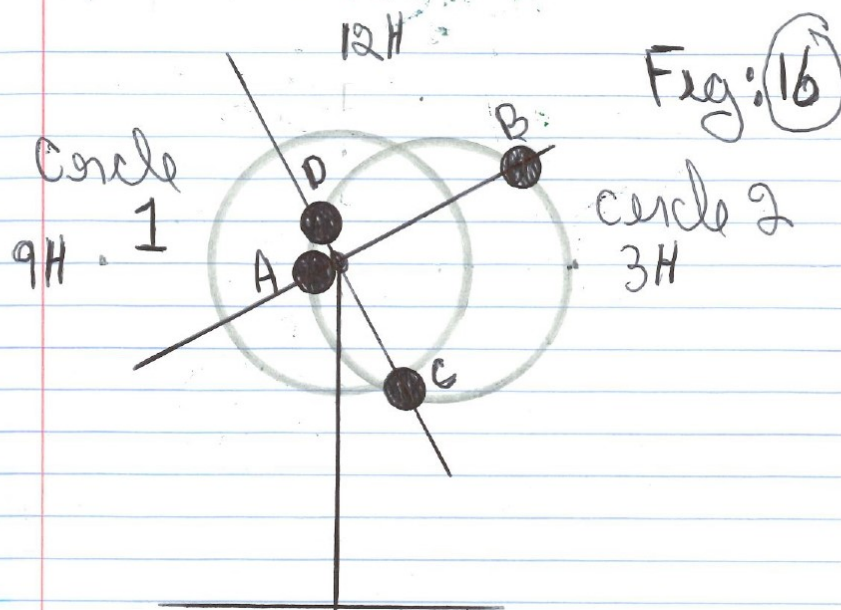
28

with only 2 axles and 4 masses, will have no choice but to rotate to the right—not only in a continual and redundant manner, but with effective torque at the hub; a kinematics that is absolutely nowhere to be found in traditional systems.

Now further down in Figure 16:

I have redrawn the same configuration from Figure 15 and I have traced a circle as explained in Figure 9, by taking the two centers of our masses at the most optimal position at 9 and 3 o'clock.

I have designated this circle as Circle 2, and then I remade the exact same circle, shifting it to the left in order to position the hub at the center of this new circle, which I called Circle 1, as illustrated below.



dans nos roues perpétuelles
 comme nos masses passeraient
 dans le Cercle no 1 pour produire
 0 couple au centre
 c'est ce qui différencie la
 cinématique d'une roue perpétuelle
 d'une roue réellement
 motrice :

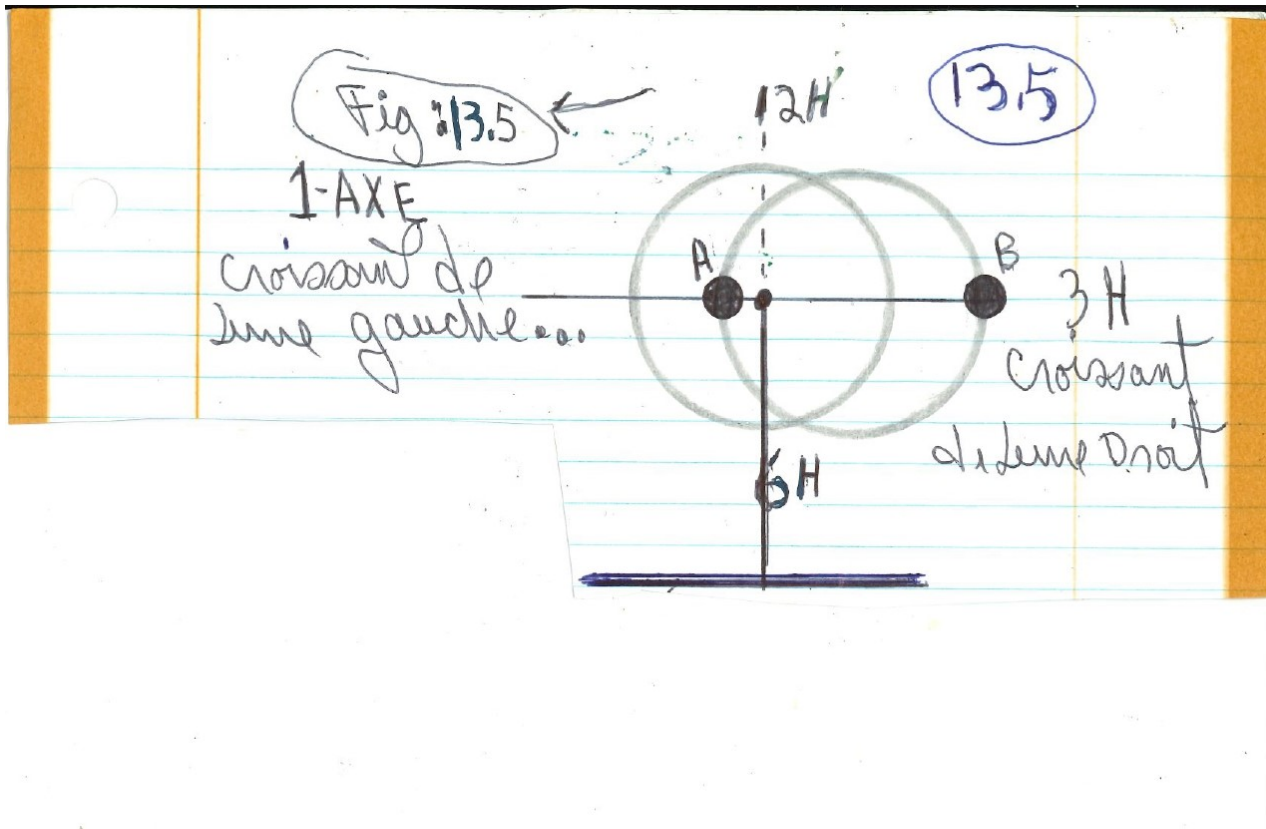
What I am saying is that, in known perpetual wheels, the masses all end up passing through the equivalent of the left circle, representing a kinematics where the left-hand mass is almost balanced relative to the hub.

This contrasts completely with the kinematics of Circle 2, where the mechanical advantage of the lever arm is obvious by bringing the left-hand mass closer to the center hub.

Furthermore, as seen in this Figure 16, such a perfectly circular kinematics of the masses begins to become functional with only two axes and 4 masses—and not 64 axles like the inspiring wheel of Aldo Costa.

With that understood, take a close look at the geometry below and the presence of two crescent moons, one on the right and the other on the left.

You can clearly perceive that on the right is where the gain begins, peaking at the horizontal 9 and 3 o'clock positions, and fades around the 27-28 minute mark, while the left crescent illustrates the exact opposite process.



The Mass Guidance Mechanism (DM) The answer is quite simple: it requires guiding the masses along a fixed, perfectly circular physical track that is offset (eccentric) relative to the central hub.

In mechanics, to force a component to follow a flawless geometric path while allowing it to continuously change its distance from the drive center (the hub), the torque transmission is decoupled from the trajectory guidance.

Here is how this concept is practically organized in reality to eliminate the kinematic disruptions at 6 and 12 o'clock:

The Stator Guide (The Eccentric Rail) Instead of leaving the masses

suspended by free articulated arms or disrupted by stoppers, a fixed circular rail (stator) is installed.

This circle is perfect and smooth.

It is not centered on the wheel's hub, but shifted (eccentric) to the right or left.

The masses are equipped with rollers or bearings that force them to strictly adhere to this predetermined diameter.2. The Drive Rotor (Sliding Arms)The central hub, which collects the driving power, features radial arms. These arms do not dictate the trajectory of the masses; they merely push or hold them during rotation.

To absorb the constant change in distance between the hub (rotor center) and the rail (circular track center), the arms must allow the masses to slide freely along flat bars.

The Mechanical Outcome of This Setup:By setting the rail's eccentricity so that the peak of the imbalance (the maximum lever arm) culminates precisely on the 9 to 3 o'clock axis, the system forces the masses:To plunge toward the hub on one side, instantly reducing the resisting lever arm.

To extend to the maximum distance on the other side, maximizing the driving lever arm.

To transition perfectly smoothly without experiencing the shock or dead center of classic wheels at 6 and 12 o'clock, since the circular path

of the rail has no disruptions.

(5)

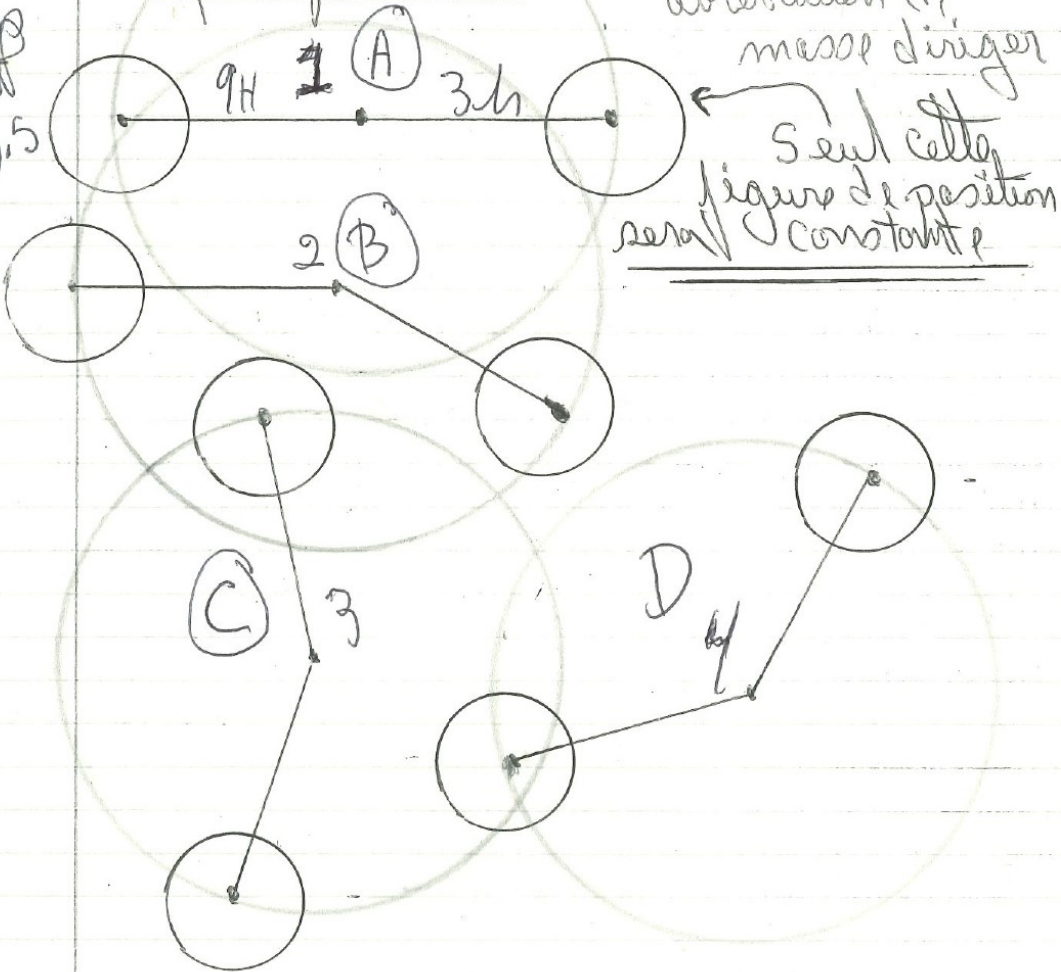
7 pages
échantillon
6.50/7.5

→ Vue de Profil Simple - principe du DM -

D-M P-5

abréviation de
masse dirigée

← Seul cette
figure de position
sera constante



ainsi 2 masses peuvent être
configurées de manière à ne
devoir passer que dans un
cercle prédéterminé

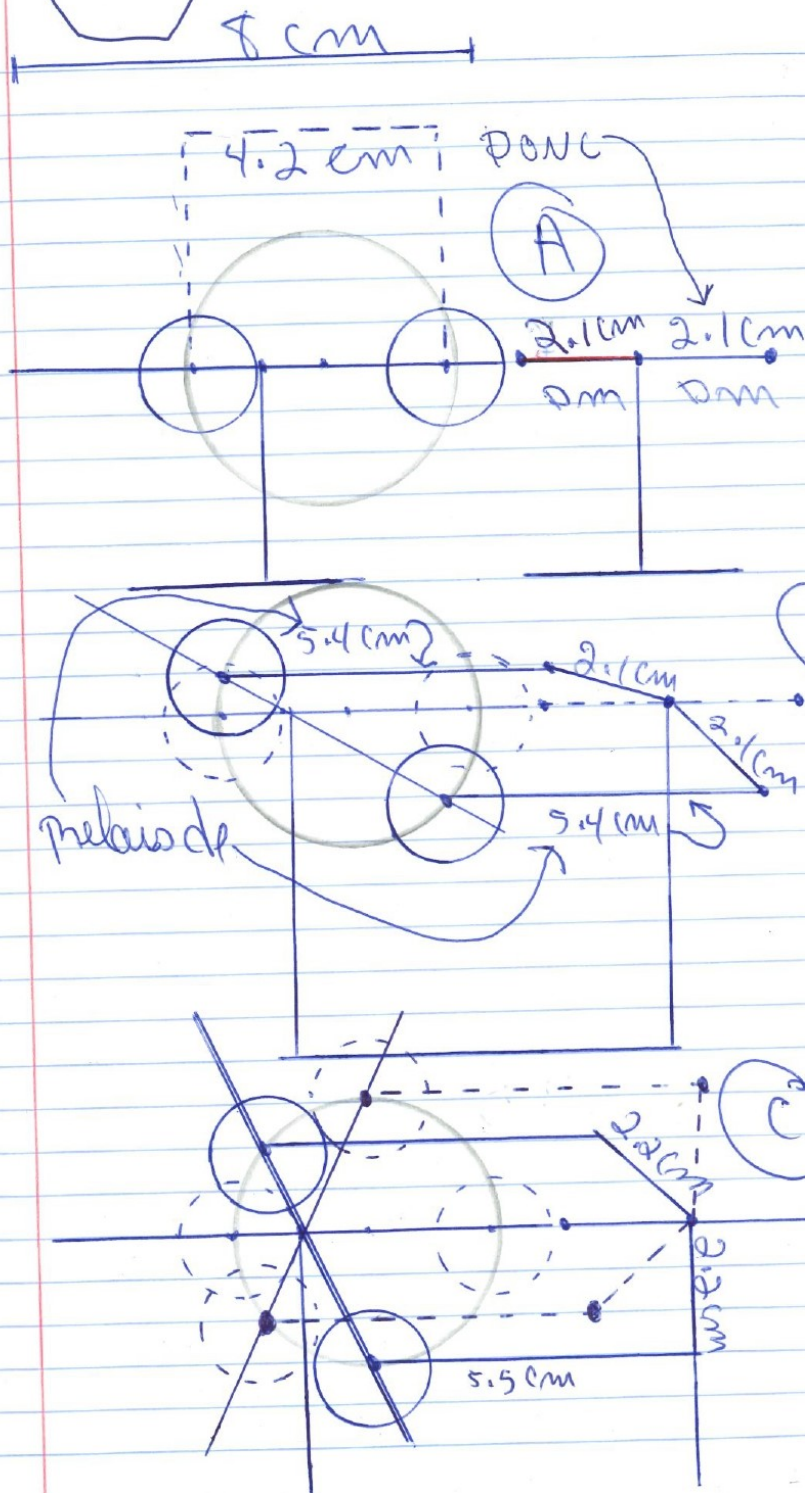
The Operating Principle of the DM Mechanism You can observe that the linkages or relays are anchored directly at the center of mass, which explains why the initial kinematic measurement must be taken from center to center of our two masses.

Let us call this simple structure a DM (D for Directing, M for Mass).

When you understand this principle, you also understand how the masses are physically constrained to move only within a predetermined circular path.

For instance, to trace a 4.2 cm circle, we simply need a 2.1 cm pivoting arm and its corresponding relay, which compensates for the distance between the position of the DM and the predetermined circle. It is exactly the same concept as using a geometric compass.

page 7 / Assemblage du DM



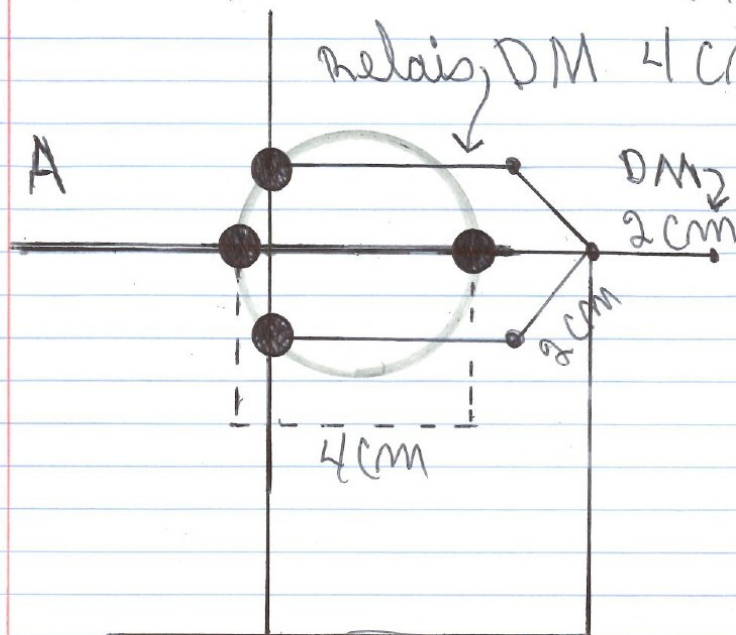
Here above in Figure C, since the design was drawn with a ruler, the physical measurement showed 5.5 cm; however, for geometric accuracy and without altering the actual data, the correct mathematical answer in Figure C is 5.4 cm, in parallel with Figure B.

Now let us analyze a two-axis configuration and return to the circular diameter through which the masses must move.

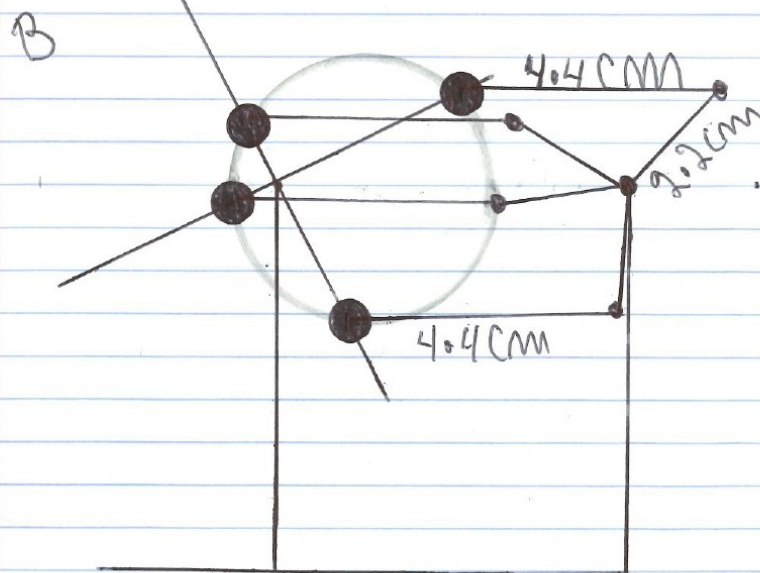
Below, we notice that if the circular diameter is, for example, 4 cm, the relay joining the arms of the DM to the center of mass will also be 4 cm.

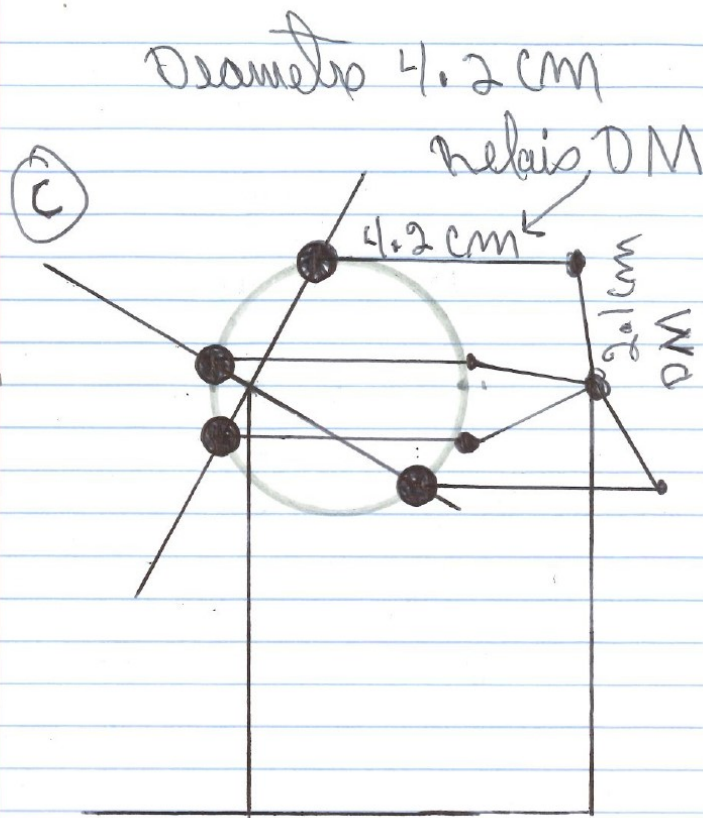
Diamètre circulaire: 4 cm

relais, DM 4 cm



ici le Diamètre est de 4.4 cm





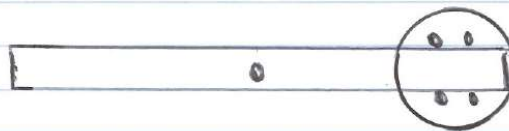
ainsi on comprend comment
contraindre 4 masses à évoluer
circulairement que dans une
trajectoire circulaire prédéterminée

It can be observed that with this kinematics, the dilemma at 6 and 12 o'clock typical of known perpetual wheels is entirely resolved thanks to the DM mechanism operating within a perfect circle. It must be specified that:

si pour tout mes précédents
 dessin j'ai simplement tracer
 en simple ligne ce qui
 est plus rapide à
 Dessiner dans la
 réalité ces lignes
 devront plutôt être
 des flats barres comme ça :



et c'est cet axe qui rentre
 dans la rondelle masse
 comme ça. Les 4 points
 sont les maubettes à l'intérieur pour



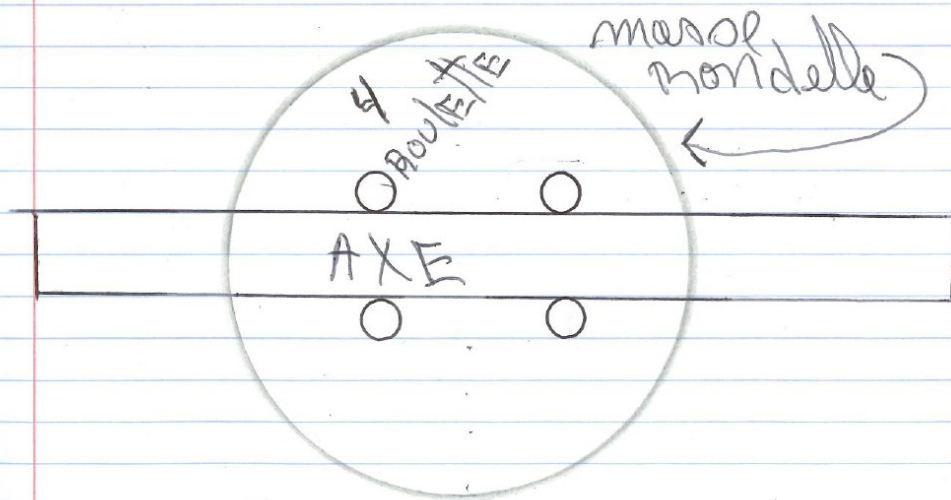
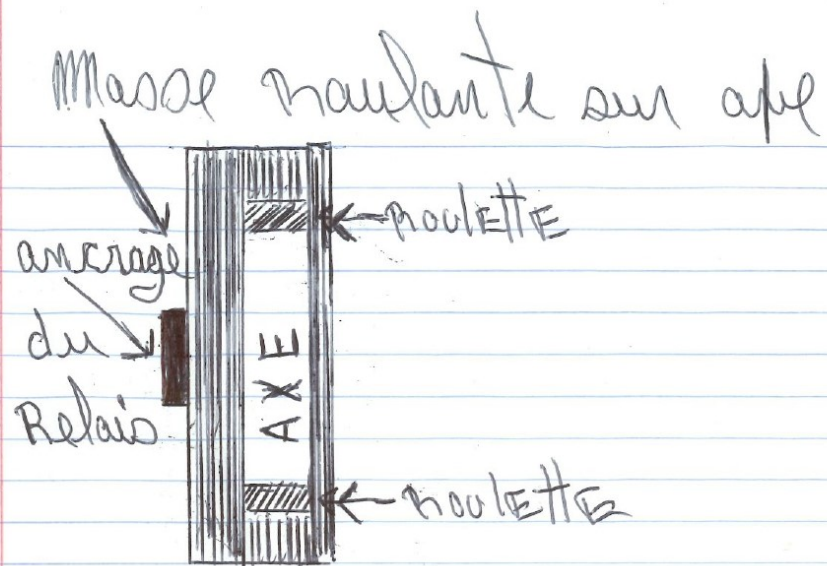
en faciliter le
 déplacement

"Even though in my previous drawings I simply drew a single line, which is faster to sketch, in reality these lines should rather be flat bars like this: and it is this axle that goes into the washer-mass like this: the 4 dots are the rollers inside to make it move easier.

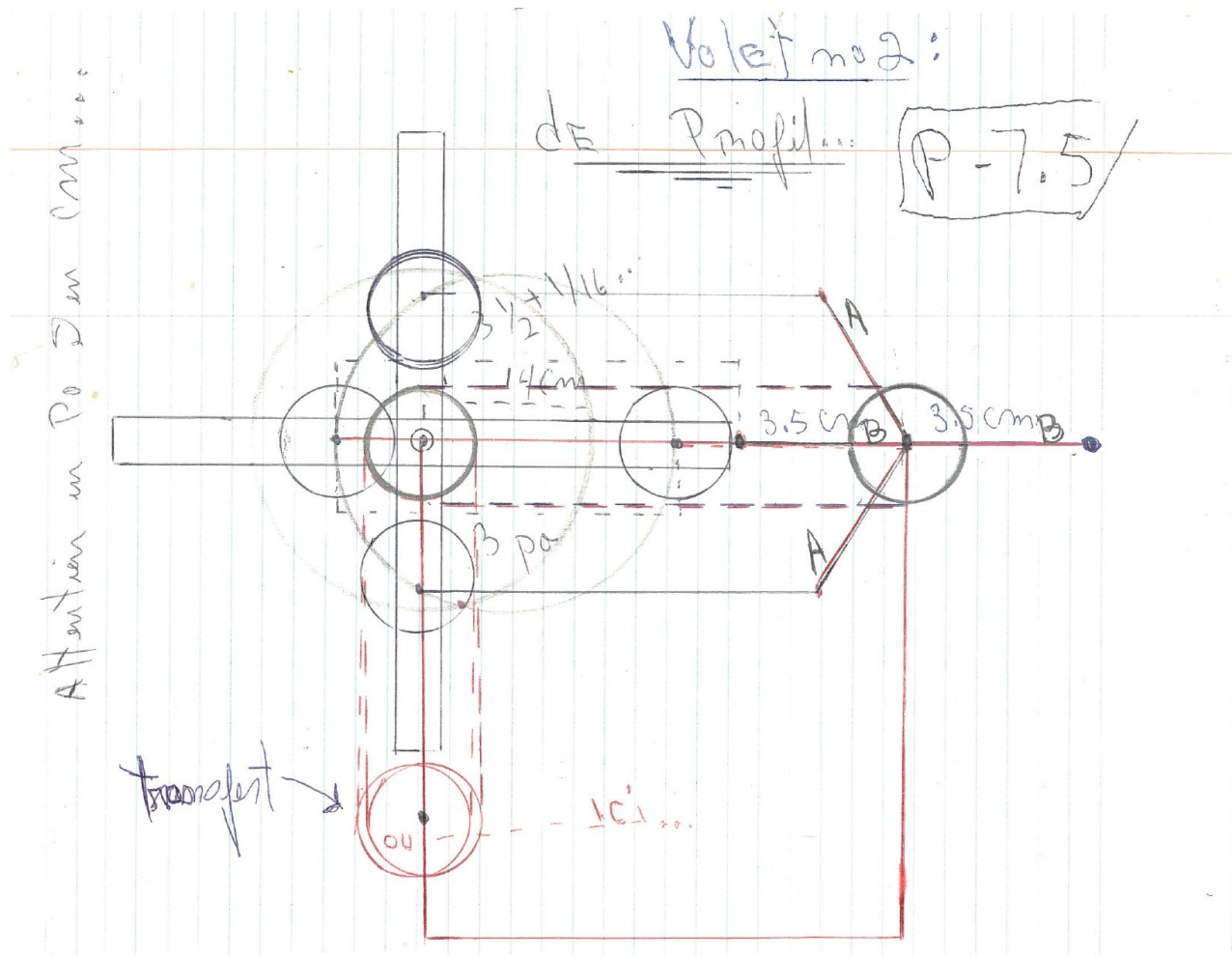
"This drawing below illustrates the rolling mass viewed in its cross-section; we can see 2 out of the 4 rollers that would be seen in profile.

We see the flat bar on which it rolls and we see the anchorage on the left for the relay at the center of this circular mass.

Since these circular masses will be gripped at their center by the relays, this demonstrates once again the reason for our initial measurement from center to center of our two masses.



Here below, this is what this kinematics with 2 axes plus the DM mechanism and flat bar axles should look like...



Up to this point, we have analyzed the system from a profile view, as it is in this view that the forces at play are best observed.

As you can see on this drawing labeled 7.5, we can notice the addition of a sprocket at the hub center and a second sprocket on each DM lever

arm.

These two sprockets, connected by a chain, force them both to move together at the same speed, making any backward movement or sudden falling of the masses—even in the critical 6 and 12 o'clock positions—entirely impossible.

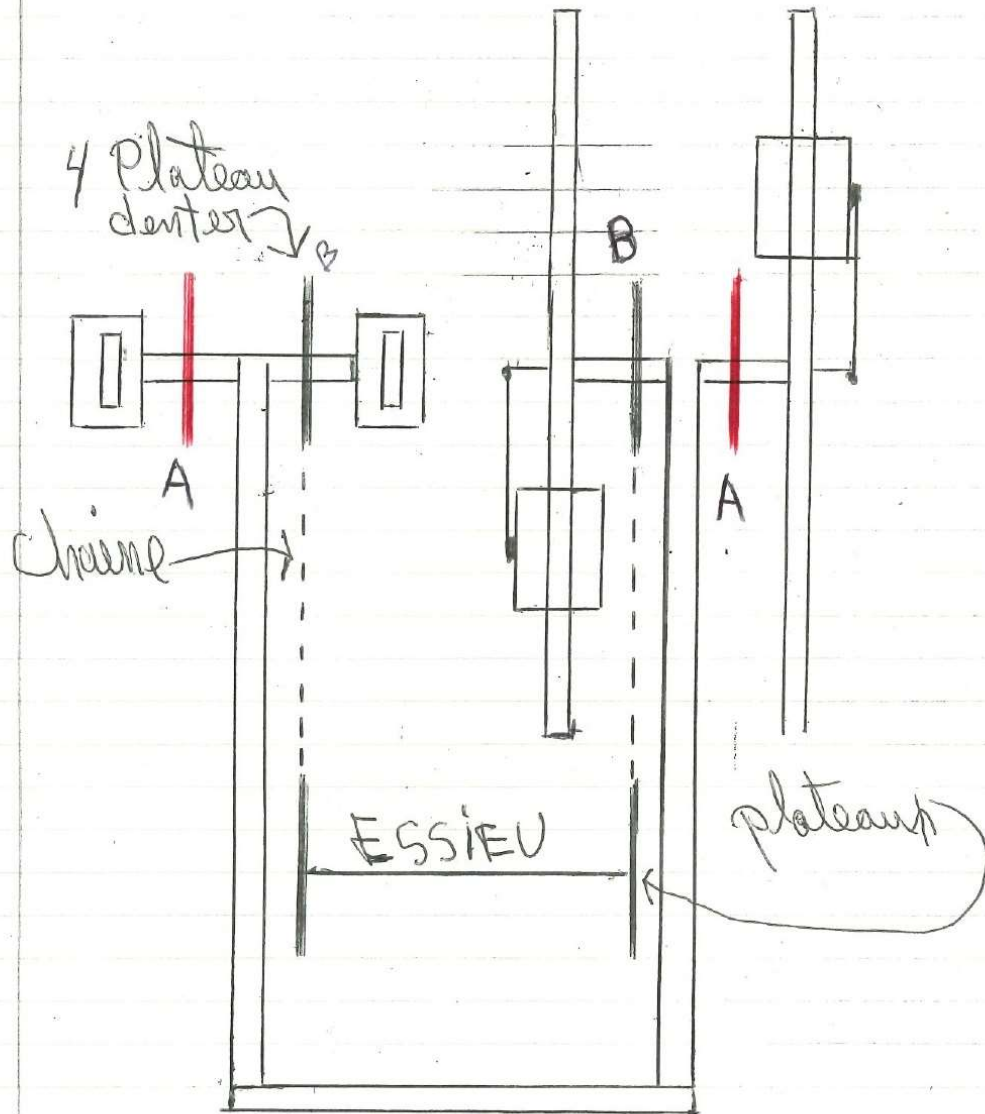
Where I wrote "transfer" (transfert) is the point where this driving force is utilized; this output could have been placed anywhere, provided that it is obviously outside the rotational diameter of the axes.

Now let us take a close look at this kinematics, but this time from the Front, to clearly see to what extent appearances are deceiving when looking from a profile view.

From a profile view, it is impossible to see this doubling of the axes, nor can we see that we have just one mass per axis.

dessin No 9

- Vue de Front



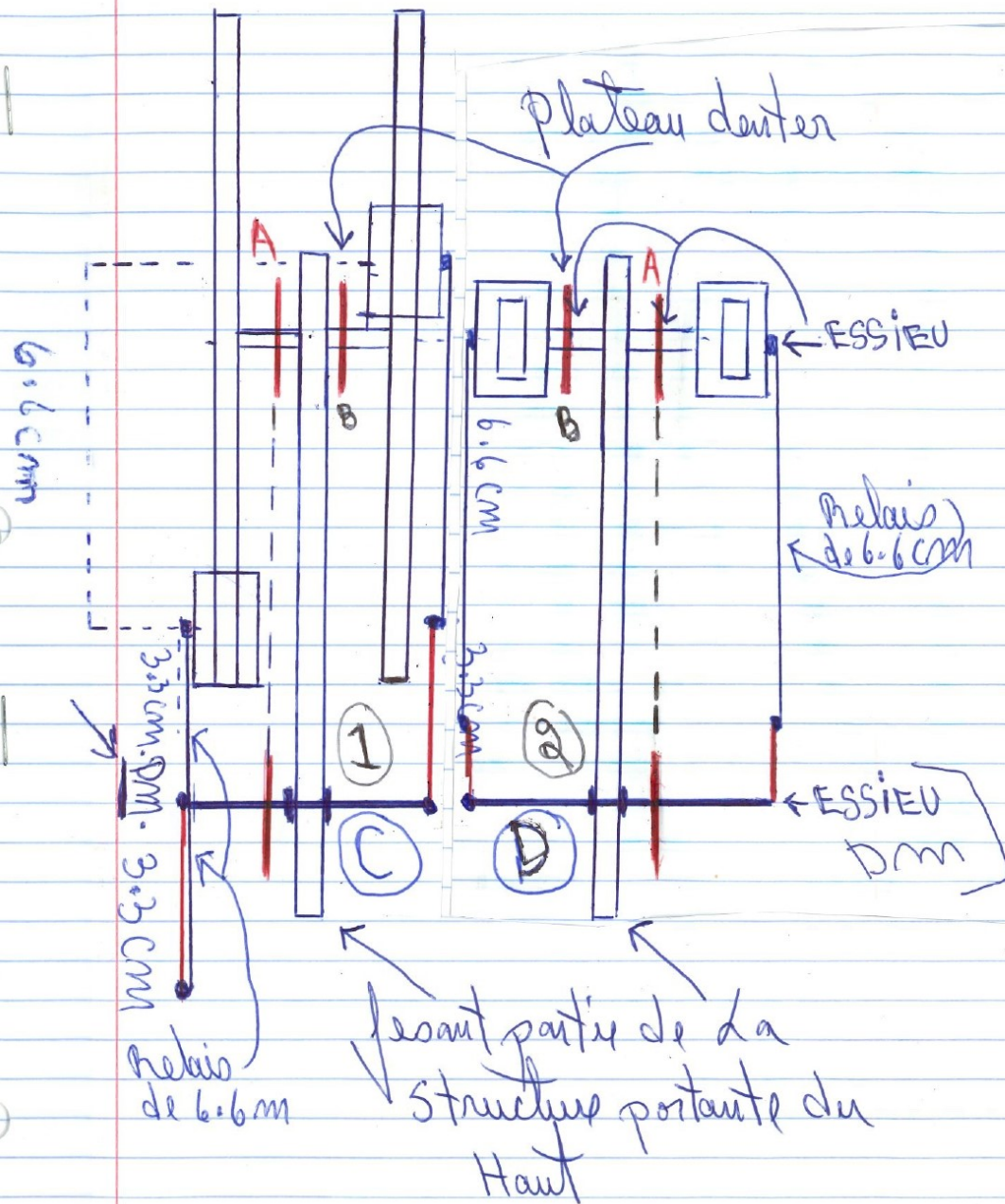
But it can be recognized that the structure is distributed at the 9, 3, 6, and 12 o'clock positions.

We can observe that the two right vertical axes are connected at the top by a sprocket to a sprocket and transfer axle lower down by a chain, and that this lower sprocket and its axle are connected to the two left horizontal axes; thus, one constrains the other.

We also see two red sprockets labeled A, which will be connected to a DM arm, as will be shown in the next drawing, this time from a ceiling view (top view).

Vue de PLAFOND

10



And the position of the sprockets will be the one seen in drawings 9 and 10, contrary to the visual impression obtained from a profile view.

Here we must clearly notice that A is connected to the DM lever arm while B (the chain), being in a vertical position as shown in drawing No. 9, is hidden from view in this ceiling view (top view).

A drawing from behind could have been made, but it would have yielded substantially the same result as the front view.

Have you ever seen such a kinematics before ?

The answer is no.

Conventional technical records only document inefficient and thermodynamically limited wheel configurations

.CONCLUSION AND INDUSTRIAL APPLICATIONS

The fundamental question raised for mechanical designers and kinematic specialists is the following:

What are the concrete applications of a rotating axle delivering high torque at low speed ?

This system proposes a geometric approach that differs significantly from conventional designs.

Although standard academic theory typically rejects such configurations by invoking the laws of thermodynamics, the precise geometry of the plans submitted here invites a direct, empirical verification in a workshop environment.

The Circular Mass Transfer System (STCM) does not belong to the category of theoretical perpetual motion systems.

It is a rigorous kinematic design developed specifically to maximize the generation of useful driving force at the central hub through eccentric guidance.

To guarantee global accessibility to this technology and prevent any retention of intellectual property, this concept is made freely available under an open-source framework.

This work is primarily addressed to professionals in mechanical fabrication: technicians, machinists, and welders who possess the practical expertise required for precision machining and structural assembly.

With the conceptual validation complete, the next phase requires the construction of a functional prototype.

The geometric tolerances indicated on the plans are clear and directly exploitable by any conventional machine shop.

The evaluation of this kinematics now relies on technical execution and load testing under real-world operating conditions.

For any request regarding technical validation, engineering exchanges, or workshop coordination related to the manufacturing of this circular mass transfer system, you may get in contact at the following address:

Benoitcc58@proton.me

Benoît Collin – Mechanical Designer – Montreal, Québec.

